

ECSE 2500

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April 24

Topic: Law of large number
Central limit theory

Recall in the
lecture:

In the limit, for n iid. RVs,

$$\lim_{n \rightarrow \infty} \text{Var}(M_n) = 0,$$

$$\lim_{n \rightarrow \infty} E[M_n] = \mu.$$

□ Weak Law of large number

Q: How fast do we get these values?

A natural way to estimate the speed is through the Chebyshev inequality:

$$P(|M_n - E[M_n]| \geq \epsilon) \leq \frac{1}{\epsilon^2} \text{Var}(M_n)$$

$$\Leftrightarrow P(|M_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2}$$

$$\text{or } P(|M_n - \mu| < \epsilon) \geq 1 - \frac{\sigma^2}{n\epsilon^2}$$

Message: This lets us bound how many observations we need to ensure that our sample mean M_n is within ϵ distance of the actual mean μ .

Say $\text{Var}(X_i) = 5$ and we want to be 99% sure that our estimate is within 0.1 of the true mean. Then we want

$$1 - \frac{6^2}{n \epsilon^2} \geq 0.99$$

$$\Rightarrow \frac{5}{n \cdot (0.1)^2} \leq 0.01 \Rightarrow n \geq \frac{5}{(0.01)^2} = 5000$$

We need at least 5000 samples so that the sample mean M_n is close to the actual mean by 0.1.

Important)

Weak law of large number:

If X_i are i.i.d., $E[X_i] = \mu < \infty$, $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|M_n - \mu| < \epsilon) = 1$$

Plain language: If we are given ϵ , we can take n large enough so that M_n is that close to the mean with high probability.

Implication: If we have a biased coin with $P(\text{head})=p$, then we can treat this as a Bernoulli RV with $P(0)=1-p$, $P(1)=p$. If we can compute M_n , then WLLN says

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - p\right| < \epsilon\right) = 1$$

M_n

Average # of success gets very close to p as n gets very large.

Limitation of WLLN: The Chebyshev inequality we use is very conservative, so the # of suggested samples/trials may be an overestimate.

Therefore, to overcome this limitation, there is also a more powerful version of WLLN; which is

□ (SLLN): Strong Law of Large Number

If X_i is a sequence of iid RVs with

$$E[X_i] = \mu < \infty, \quad \text{Var}(X_i) = \sigma^2 < \infty$$

then $P\left(\lim_{n \rightarrow \infty} M_n = \mu\right) = 1$.

The difference between WLLN and SLLN is a bit subtle. It basically says that any sequence of sample means must eventually converge to the actual mean.

☐ Central limit theorem

The WLLN and SLLN talk about how the sample mean converges to the actual mean.

The central limit theorem is stronger: It talks about how the PDF/CPF of the sample mean M_n converges. The result is that no matter the incoming distribution of X_i , the limiting distribution of M_n is (in certain sense)

Gaussian !

Recall $S_n = \sum_{i=1}^n X_i$; $E[S_n] = n\mu$, $\text{Var}(S_n) = n\sigma^2$
 $M_n = \frac{1}{n} \sum_{i=1}^n X_i$; $E[M_n] = \mu$, $\text{Var}(M_n) = \frac{\sigma^2}{n}$

If we define a new RV Z_n :

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}$$

$$E[Z_n] = \frac{1}{\sigma\sqrt{n}} (E[S_n] - n\mu) = 0$$

$$\text{Var}(Z_n) = \frac{1}{\sigma^2 n} \text{Var}(S_n) = 1$$

At each value n , Z_n is a zero-mean, unit variance random variable.

The Central limit theorem states that

$$\lim_{n \rightarrow \infty} P(Z_n \leq z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-\frac{x^2}{2}} dx$$

$$\Rightarrow \lim_{n \rightarrow \infty} F_{Z_n}(z) = \phi(z) = 1 - Q(z)$$

↑
CDF of standard Gaussian RV (mean 0, variance 1)

That is, the CDF of Z_n converges to the Gaussian CDF for any value z , as $n \rightarrow \infty$.

To rigorously prove CLT, we need more advanced tools in probability, but you can easily use numerical software to verify this result.

Implication of CLT - When n is big enough, we can use CLT to get a good approximation of the probabilities of events by creating a corresponding Gaussian RV and looking at the Q table.

More specifically, CLT says that

$$P(Z_n \leq z) \approx \Phi(z)$$

$$\Rightarrow P\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq z\right) \approx \Phi(z)$$

$$\Rightarrow P\left(\frac{nM_n - n\mu}{\sigma\sqrt{n}} \leq z\right) \approx \Phi(z)$$

$$\begin{aligned} \Rightarrow P\left(|M_n - \mu| \leq \frac{z\sigma}{\sqrt{n}}\right) &\approx \Phi(z) \\ &\approx 1 - 2Q(z) \end{aligned}$$

Example

Suppose $X_1, X_2, \dots, X_n$ are iid RV with Bernoulli distribution, i.e.,

$$P(X_i=1) = P(X_i=0) = \frac{1}{2}$$

We have $E[X_i] = \frac{1}{2}$, $\text{Var}(X_i) = \frac{1}{4} = \sigma^2$

Q: How close is M_n to $\frac{1}{2}$?

Say $n = 10000$ and $z = 1$

$$P\left(|M_n - \mu| \leq \frac{z\sigma}{\sqrt{n}}\right) \approx 1 - 2\phi(z)$$

$$\Rightarrow P\left(|M_n - \frac{1}{2}| \leq \frac{1}{200}\right) \approx 1 - 2\phi(1)$$

check Q-table
= 0.682

Q: For what value of n , we will have

$$P(|M_n - \frac{1}{2}| \leq 0.005) \geq 0.9 ?$$

$$\Rightarrow 1 - 2\phi(z) \geq 0.9, \quad \phi(z) \leq 0.05$$

Check Q-table, $z \geq 1.65$

So it means that

$$\frac{Z_6}{\sqrt{n}} = 0.005$$

$$\Rightarrow \frac{1.65 \times \frac{1}{2}}{\sqrt{n}} = 0.005 \Rightarrow \sqrt{n} = 165$$
$$\Rightarrow n = 27225$$

Note: When n is big enough that the CDF of Z_n approximates Gaussian, then the CLT will give better results than standard concentration inequalities (e.g., Markov, Chebyshev or Chernoff), but when n is small, the Gaussian approximation in CLT may not be accurate!

Exam 3

May 4 noon - 1:30 PM Exam 3

Two pages ~~two-sided~~ notes

6-7 problems, ~ 2 prob from HWS

2 ~ material before Exam 1, 2 ~ related Exam 2, 2-3 after Exam 2.